Module 4 Air Sampling | Q20. | 10/26/2026 | Professor Rodriguez

Sorbent tube to ppm

- A charcoal tube samples Toluene ($C_6H_5CH_3$) for 240 min.
  (MW: 92.14 g/mol)
- Pre-flow: 0.050 L/min
- Post flow: 0.0490 L/min

① $\left(\frac{0.050\ \text{L/min} + 0.0490\ \text{L/min}}{2}\right) = 0.0495\ \text{L/min}$

- The lab reports 1.15 mg. check the flow, then calculate the air volume, the concentration in mg/m³ and the concentration in ppm.

② "Time" → 240 min $\left(\frac{0.0495\ \text{L}}{\text{min}}\right)$ = 11.88 L ("flow" → "volume")

③ "density" (mass/volume) = [concentration] [1.15 mg / 11.88 L] ("concentration")

"mass" → 1.15 mg
"volume" → 11.88 L

$\left(\frac{1.15\ \text{mg}}{11.88\ \text{L}}\right)\left(\frac{1000\ \text{L}}{1\ \text{m}^3}\right) = [96.801\ \text{mg/m}^3]$

"unit conversion"

$\frac{1.15\ \text{mg}}{11.88\ \text{L}} = [0.096800\ \text{mg/L}]\left(\frac{1000\ \text{L}}{1\ \text{m}^3}\right) = [96.801\ \text{mg/m}^3]$

Part 2

→ Toluene: mw = 92.14 g/mol

→ Pre-flow: 0.050 L/min
→ Post flow: 0.0490 L/min
} → Avg. flow: 0.0495 L/min

→ Mass: 1.15 mg

→ Time: 240 min

→ Volume: 11.88 L

→ [Concentration]: [96.801 mg/m³]

Conversion formula — Derived from the ideal gas law

$$[ppm_v] = [mg/m^3]\left[\frac{RT}{P\,Mr}\right]$$

(273.15 K)

(760 mmHg)

$$R = \left(62.4\ \frac{ml \times mmHg}{mmol \times K}\right)$$

Toluene: mw = 92.14 g/mol → 92.14 mg/mmole

"R-constant" — Kelvin

$$[ppm_v] = [96.80\ mg/m^3]\left[\frac{\left(62.4\ \frac{ml \times mmHg}{mmol \times K}\right)(273.15K)}{(760\ mmHg)\left(92.14\ \frac{mg}{mmole}\right)}\right]$$

"mercury" — mg/mmole

$$[ppm_v] = [96.80\ mg/m^3]\left[\frac{17044.56}{7002 6.4}\right]$$

$$[ppm_v] = 23.56$$